

VC BIRD INTL, Antigua and Barbuda

WMO: 788620

Lat: 17.137N

Lon: 61.793W

Elev: 62

StdP: 14.66

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	71.2	72.0	61.0	80.2	76.5	62.8	85.6	76.9	25.5	77.9	23.3	78.5	4.2	60	0.507

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.3	89.1	79.1	88.1	78.8	87.7	78.6	80.8	86.5	80.0	85.9	79.6	85.6	14.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.1	151.3	84.8	78.5	148.3	84.4	77.4	142.8	83.8	44.5	86.6	43.6	86.2	43.1	85.9	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.4	20.7	19.1	DB	68.0	90.7	1.5	1.2	67.0	91.6	66.1	92.2	65.3	92.9	64.2	93.7	
			WB	63.3	82.2	1.7	1.8	62.1	83.4	61.1	84.5	60.2	85.4	58.9	86.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	80.7	78.0	77.9	78.5	79.6	81.0	82.4	82.9	83.2	82.7	82.1	80.7	79.1
	DBStd	2.47	1.42	1.38	1.58	1.53	1.74	1.51	1.48	1.44	1.66	1.73	1.73	1.62
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11202	869	781	884	887	961	972	1020	1030	980	995	922	901
	CDD65	5727	404	361	419	437	496	522	555	565	530	530	472	436
	CDH74	56632	2873	2471	3154	3819	5011	5837	6445	6705	6143	5855	4675	3644
CDH80	14628	273	262	445	703	1198	1714	2112	2381	2115	1824	1097	504	
Wind	WSAvg	11.8	12.3	12.7	12.1	12.3	12.0	13.0	13.9	12.4	9.8	9.3	9.8	11.5
Precipitation	PrecAvg	46.3	2.6	1.6	1.7	2.8	3.4	3.8	4.6	5.6	5.7	5.8	5.3	3.4
	PrecMax	71.8	5.3	3.3	4.1	9.9	9.7	8.0	9.7	10.2	12.7	10.6	12.6	9.1
	PrecMin	30.2	1.1	0.2	0.4	0.6	0.2	1.1	1.0	2.9	1.7	2.0	1.4	1.4
	PrecStd	9.7	0.9	0.6	0.8	1.9	2.5	1.8	2.0	1.9	2.5	2.6	3.3	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.5	85.1	86.1	86.4	88.5	88.2	89.7	89.8	89.8	89.6	88.0	86.0
		MCWB	74.6	73.9	74.8	75.6	77.5	78.4	79.8	80.3	78.9	79.1	78.3	77.7
	2%	DB	82.8	83.4	84.4	85.7	87.0	87.6	88.0	88.2	88.2	88.0	86.4	84.4
		MCWB	73.8	73.1	74.1	75.3	77.2	78.2	78.9	79.3	78.9	78.8	77.7	75.7
	5%	DB	82.5	82.5	83.3	84.4	86.0	86.4	87.3	87.8	87.9	87.3	85.8	83.8
		MCWB	73.5	72.8	73.5	74.9	76.9	77.6	78.5	79.1	78.9	78.5	77.4	75.2
	10%	DB	81.8	81.8	82.5	83.9	84.6	86.0	86.2	86.9	87.0	86.2	84.5	82.6
		MCWB	73.2	72.5	73.1	74.6	76.2	77.4	77.9	78.6	78.6	78.1	76.6	74.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.3	76.1	77.1	78.2	79.7	80.6	81.8	82.3	80.9	80.7	79.9	79.3
		MCDB	81.9	82.2	82.6	83.7	85.8	86.3	87.1	87.1	87.3	87.1	85.3	83.4
	2%	WB	75.8	74.7	75.8	77.0	78.5	79.6	80.2	80.8	80.1	79.9	79.0	77.8
		MCDB	81.2	81.0	81.8	83.1	84.6	85.5	85.7	86.4	86.4	86.1	84.6	82.4
	5%	WB	74.7	74.1	74.8	76.2	77.9	78.8	79.6	80.0	79.7	79.2	78.2	76.6
		MCDB	80.5	80.3	81.1	82.3	84.0	84.9	85.2	85.7	86.0	85.3	83.9	81.6
	10%	WB	74.0	73.4	74.1	75.5	77.2	78.1	78.9	79.4	79.1	78.7	77.6	75.7
		MCDB	79.9	79.6	80.5	81.6	83.3	84.2	84.7	85.3	85.4	84.6	83.3	81.0
Mean Daily Temperature Range	5% DB	MDBR	8.1	8.7	8.8	8.7	8.3	7.6	7.8	8.3	9.0	9.0	8.7	8.1
		MCDBR	8.5	9.3	9.4	9.5	9.3	7.9	8.2	8.7	9.7	9.8	9.4	8.6
		MCWBR	4.6	4.5	4.5	4.4	4.2	4.0	4.1	4.3	4.5	4.8	4.8	4.7
	5% WB	MCDBR	8.1	9.0	9.2	9.1	8.7	7.8	8.1	8.6	9.1	9.2	9.1	8.2
MCWBR		4.9	4.7	4.8	4.8	4.5	4.4	4.7	4.7	4.6	5.0	5.0	5.0	
Clear-Sky Solar Irradiance	taub	0.363	0.370	0.371	0.403	0.442	0.520	0.512	0.476	0.432	0.389	0.372	0.369	
	taud	2.603	2.575	2.571	2.503	2.407	2.134	2.157	2.300	2.464	2.602	2.639	2.616	
	Ebn at Noon	294	297	298	287	272	249	252	263	276	287	289	289	
	Edh at Noon	30	32	33	35	38	49	49	43	36	30	28	29	
All-Sky Solar Radiation	RadAvg	1579	1804	2022	2094	2027	2035	2040	2032	1892	1697	1533	1470	
	RadStd	53	47	58	83	113	75	100	68	90	100	95	64	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	-0.50	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page